Dyllon A. Dunton | Pittsfield, Maine | (207) 313-9017 | duntondyllon@gmail.com
Robotics & AI Engineer (M.S. Candidate)
Portfolio: dyllondunton1.github.io
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Professional Summary

Master's student in Computer Engineering specializing in robotics and applied machine learning, with dual bachelor's degrees in Electrical and Computer Engineering. Experienced in designing physics-informed ML systems, latent-state models, and simulation-driven AI pipelines for perception and prediction tasks across research and industry settings.

Education

University of Maine – Orono, ME
• M.S. in Computer Engineering, Robotics & AI Focus, GPA: 4.0 (Expected May 2026)
• B.S. in Computer Engineering & B.S. in Electrical Engineering, GPA: 3.748 (May 2024)
• Robotics Team Officer (Software/Hardware Co-lead, Treasurer) (2022-2024)

Experienced Technical Skills

• **ML/AI:** PyTorch, TensorFlow, Latent Diffusion Models, Physics-Informed Neural Networks (PINNs), System Identification (SINDy), Reinforcement Learning (DDQN), Retrieval-Augmented Generation (RAG)
• **Programming:** Python, C++, C, Java
• **Robotics/Systems:** ROS 2, Docker, Arduino, Raspberry Pi, Motor Control, Sensor Integration
• **Simulation/Modeling:** MATLAB, Time-series modeling, Reduced-order Modeling
• **Data/Infrastructure:** SQL (MySQL, PostgreSQL, Oracle)

Relevant Experience

Graduate Research Assistant – UMaine ECE Dept. | *Jul 2024 – Present*
• Designed and implemented a physics-informed autoencoder for high-dimensional offshore wind turbine motion, producing stable latent representations for downstream predictive modeling
• Integrated temporal, spectral, and physical consistency losses and validated latent representations using reconstruction error, spectral fidelity, and rollout behavior under simulated operating conditions *(manuscript under peer review)*
• Built a latent diffusion inference pipeline enabling fast sampling from compressed state representations
• Developing a latent-space predictive model to forecast structural motion, with current focus on long-horizon rollout stability

Engineering Intern – General Electric Gas Power | *Mar 2021 – May 2024*
• Developed an SQL-based data retrieval and reporting system to support real-time decision-making on a production inspection line
• Programmed embedded sensing, control, and logging systems for an in-house manufacturing process, operating under high throughput production reliability constraints

Research Assistant – Advanced Structures & Composites Center | *Aug 2022 – Dec 2023*
• Developed a ROS2/Python calibration system for a multi-camera FLIR Boson array tracking 3D-print progress and failure detection
• Containerized calibration and perception workflows using Docker for reproducibility